AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 7 · STUDENT LESSON

Sampling and Intended Use

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Define the use. Count the conditions. Name the gap.

Compare source-aware sample coverage without treating volume as accuracy.

Keep one habit: Intended use → complete A/B manifests → four-cell counts → twelve-plant picture → collection lab → independent outdoor plan.

MAIN LESSON · SECTION 1

Who and what must this model work for?

> Compare the intended conditions with the actual sample before making a performance claim.

Current sampling goals

I can define the population and conditions in an intended-use statement.

I can compare sample coverage using explicit condition counts and source IDs.

I can plan collection and evaluation for missing conditions without claiming guaranteed accuracy.

Retrieve labeling and source reasoning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.
- Why can two annotators agree while both apply the wrong label rule? What information about collection conditions belongs with those labels?

Your end goal

By the end, I can explain sampling and **intended use** using a traced example and a stated limitation.

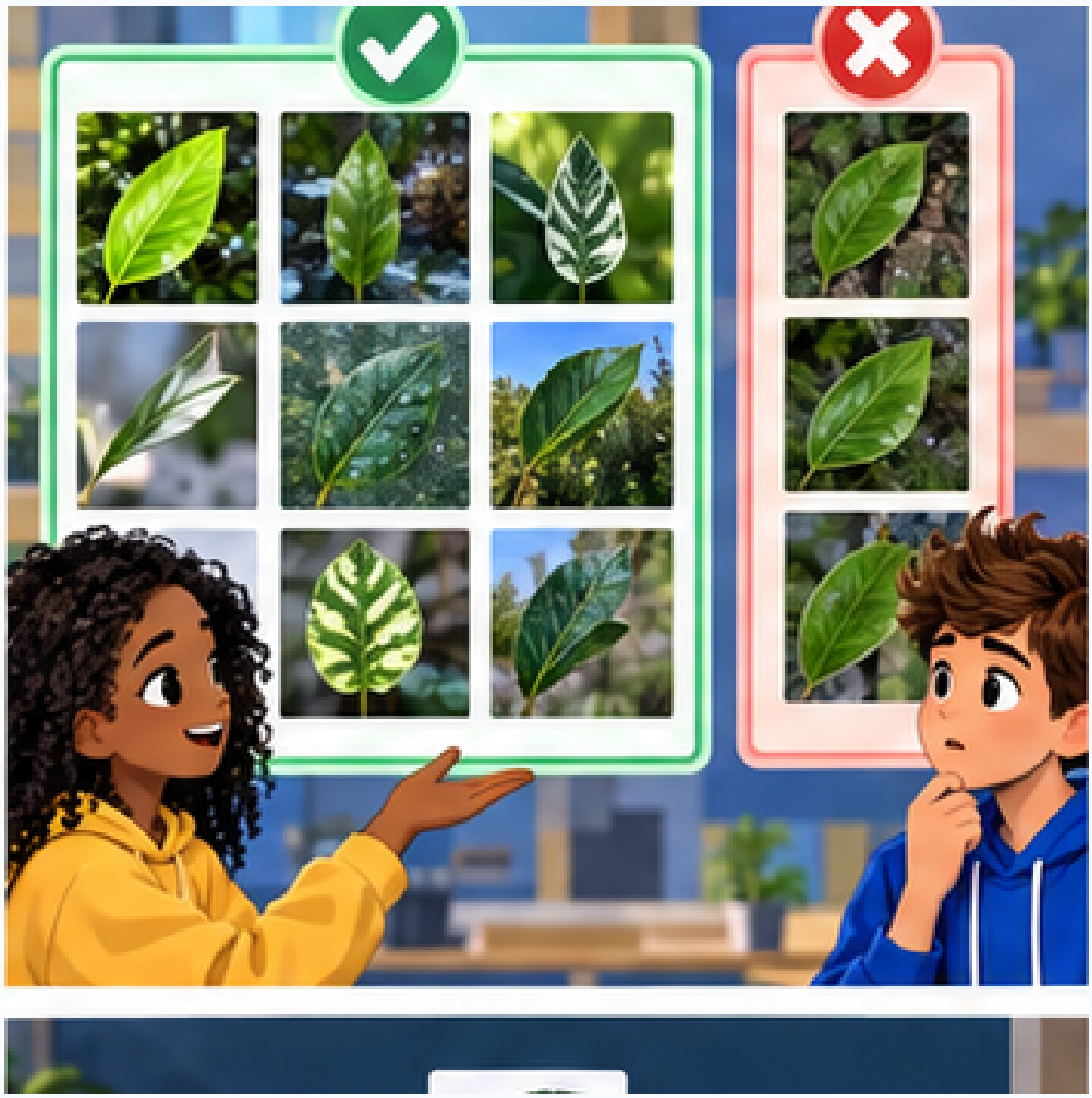
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Training examples should represent the people, objects, conditions, and devices described by the intended-use statement.



LEARNING TARGET **Training** uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Read two complete sample manifests against one declared use

> Sample size and condition coverage answer different questions.

Intended-use contract

A sample is the examples collected; the intended population is the set of objects and conditions the proposed use concerns. Define the use before choosing a convenient table. This fictional classroom wants plant-photo classification for two light settings (day and evening) and two cameras (A and B), for the declared toy label rule.

The four combinations all matter here; equal counts are a classroom comparison choice, not a universal requirement for every real deployment.

Complete photo manifests

Sample	Plant IDs / condition	Photo count
A	P01–P09: day, camera A; one photo per plant	9
A	P10: evening, camera A	1
B	Q01–Q03: day, camera A	3
B	Q04–Q06: day, camera B	3
B	Q07–Q09: evening, camera A	3
B	Q10–Q12: evening, camera B	3

Source, row meaning and unknown dimensions

Every row here is a compact manifest entry, not one photo: expand the named ID range to recover each individual photo. All IDs are distinct within each sample, all data are invented and no model outputs are supplied. Source is the teacher-created Coverage v1 fixture.

Intended use is classroom comparison only; real navigation, diagnosis and broad plant-health claims have no evidence here.

Record plant type and label-class coverage as unknown because this fixture does not supply them.

▼ Review · Earlier trip-inspection source

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

Trip ID	Duration in minutes	What is visible
A	8	A recorded duration
B	None	An unknown duration
B	None	The same ID appears again
C	-2	A negative duration
D	12	A recorded duration

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use sampling and **intended use** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Count each intended condition and each distinct source plant

> Coverage is an input-data description, not a model score.

Complete counting program

```
samples = {
    "A": [(f"P{i:02}", "day", "A") for i in range(1, 10)]
         + [("P10", "evening", "A")],
    "B": [(f"Q{i:02}", light, camera)
          for start, light, camera in [(1, "day", "A"), (4, "day", "B"),
                                       (7, "evening", "A"), (10, "evening", "B")]
          for i in range(start, start + 3)]
}
for name, rows in samples.items():
    print(name, len(rows), len({plant for plant, light, camera in rows}))
    for light, camera in [("day", "A"), ("day", "B"),
                          ("evening", "A"), ("evening", "B")]:
        count = sum(l == light and c == camera for plant, l, c in rows)
        print(light, camera, count)
```

Trace every condition count

Condition	Sample A	Sample B
Day/camera A	9	3
Day/camera B	0	3

Condition	Sample A	Sample B
Evening/camera A	1	3
Evening/camera B	0	3
Total photos / distinct plants	10 / 10	12 / 12

Explain the bounded comparison

A is 9/10 daylight and contains no camera-B photos. B covers all four stated combinations with three photos each. That supports a narrower claim about these recorded conditions, not greater accuracy or full representativeness: species, labels, selection method and real intended-use frequencies are not supplied.

The program assumes valid nonempty tuples and correct condition strings; it counts records, not measurement truth. Different views of the same plant are related observations and must retain the shared plant ID for later split design.

▼ Review · Earlier trip audit program and assumptions

```
rows = [("A", 8), ("B", None), ("B", None),
        ("C", -2), ("D", 12)]
ids = set()
missing = 0
negative = 0
for trip_id, minutes in rows:
    ids.add(trip_id)
    if minutes is None:
        missing = missing + 1
    elif minutes < 0:
        negative = negative + 1
print(len(rows), len(ids), missing, negative)
```

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

Turn an observation into a documented action

FINDING	WHAT IT ESTABLISHES	NEXT ACTION
B appears twice	Identity needs review	Check export history
Two None values	2 of 5 rows lack duration	Keep unknown explicit
C has minus 2	Invalid elapsed duration	Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

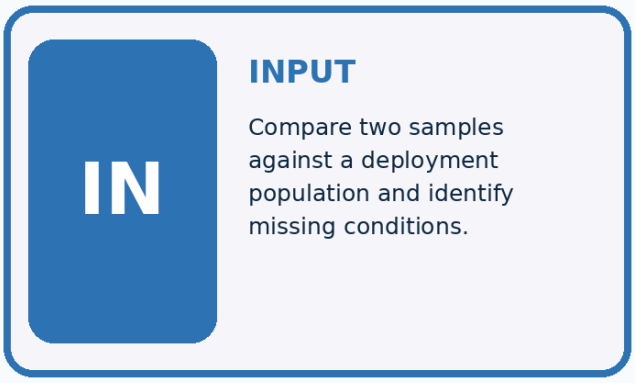
A successful run does not certify data quality.

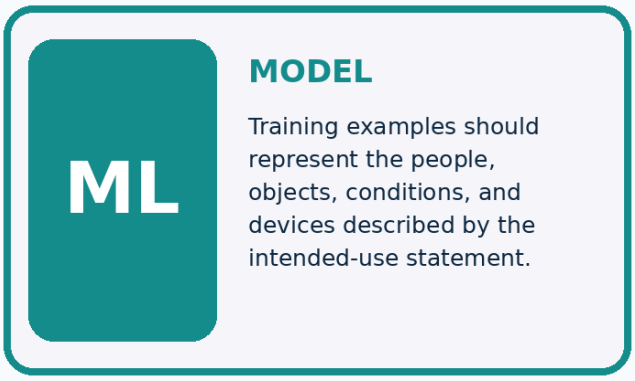
▼ **Earlier wording, alternatives and picture transcripts**

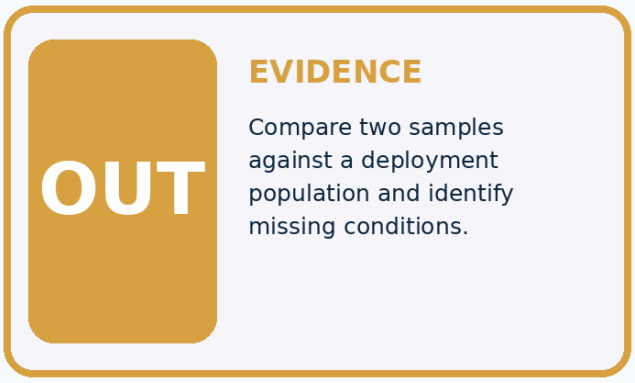
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Compare two **sample**s against a deployment **population** and identify missing conditions.

MODEL IDEA **Training** examples should represent the people, objects, conditions, and devices described by the intended-use statement.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Intended use** includes indoor classroom plants under day and evening light. **Sample** A has 90% daylight photos from one camera.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

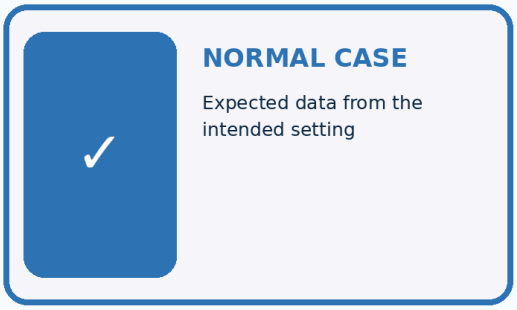
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

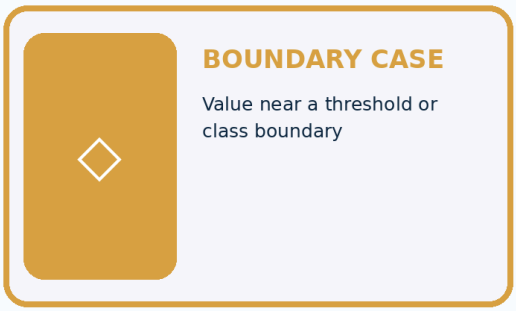
Compare a bigger familiar sample with missing intended conditions

> More rows do not necessarily add new conditions or independent plants.

Four evaluation conditions



NORMAL Expected data and expected behavior.

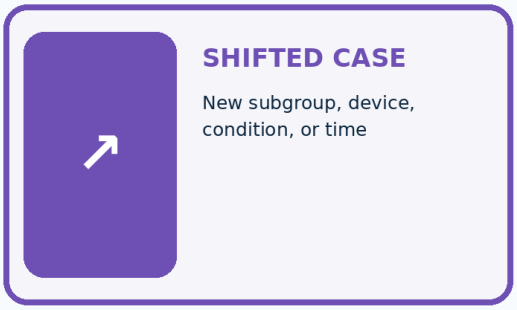


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Collect 12 more midday windowsill photos of the same plants.

▼ **Check after predicting**

Prediction: The evening-coverage gap remains.

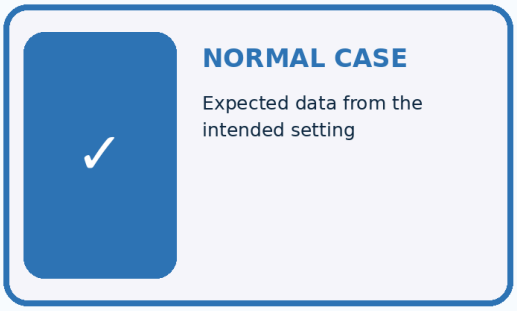
Reason: More rows in the same condition do not supply **evidence** about the missing condition.

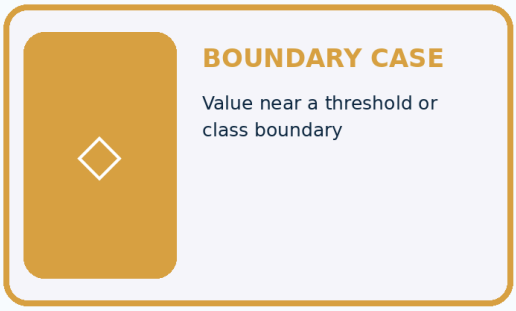
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

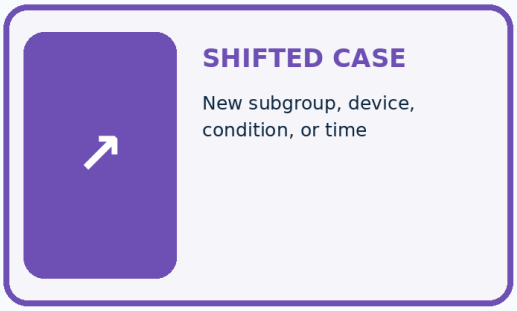
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the twelve-plant picture without confusing photos and plants

> A missing evening condition stays missing after midday duplication.

Read the pictured facts and counting scope

Sample for the conditions in which the plant classifier will actually be used.

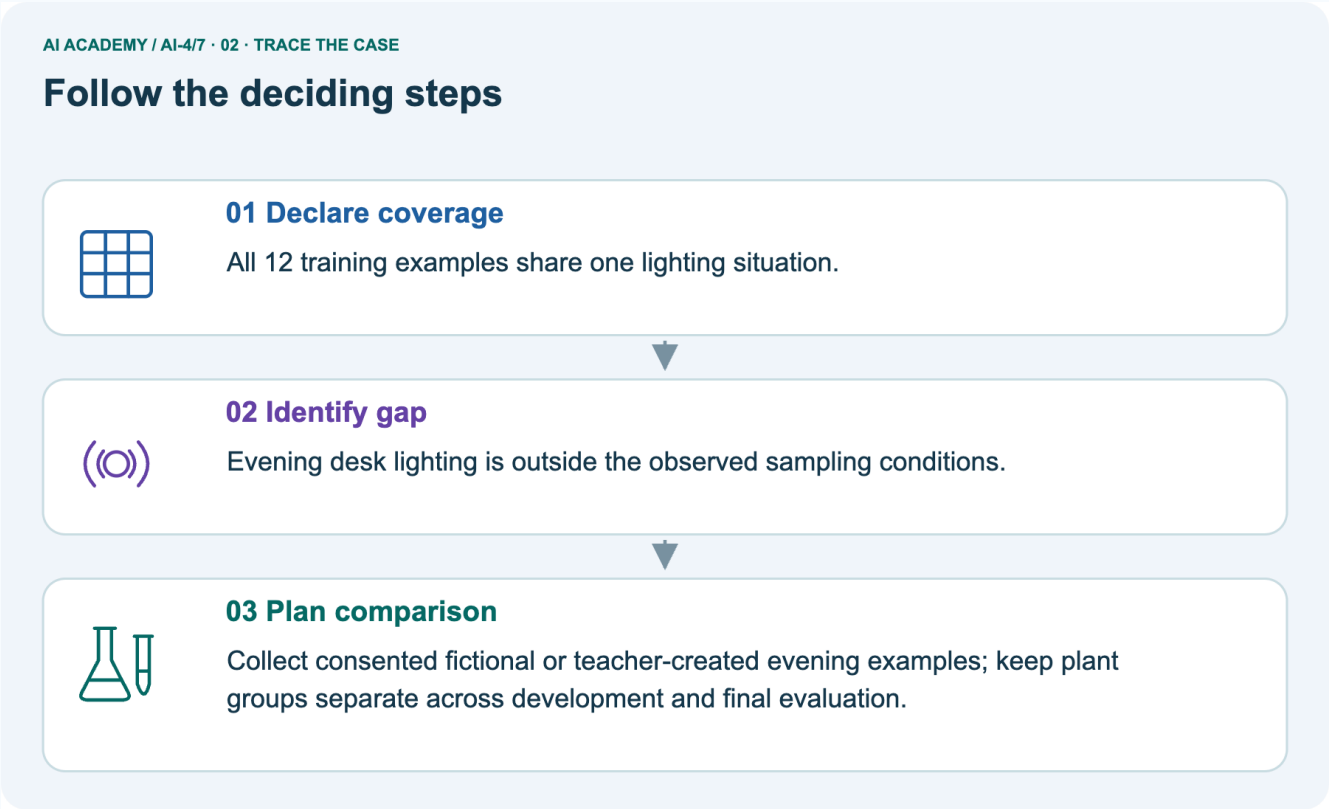
Training set: 12 windowsill plants photographed at midday. **Intended use** includes evening desk photos; no evening examples have been collected.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate twelve-plant example: all initial observations are midday windowsill photos and evening desk use is intended. Adding twelve new views of those same plants at midday gives24 photos but still12 distinct plants and no evening observations.

A collection plan should include the missing condition and record source groups; no exact performance change or guaranteed sufficient sample size follows from these counts.

Trace coverage and collection plan



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare coverage	All 12 training examples share one lighting situation.
2. Identify gap	Evening desk lighting is outside the observed sampling conditions.
3. Plan comparison	Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

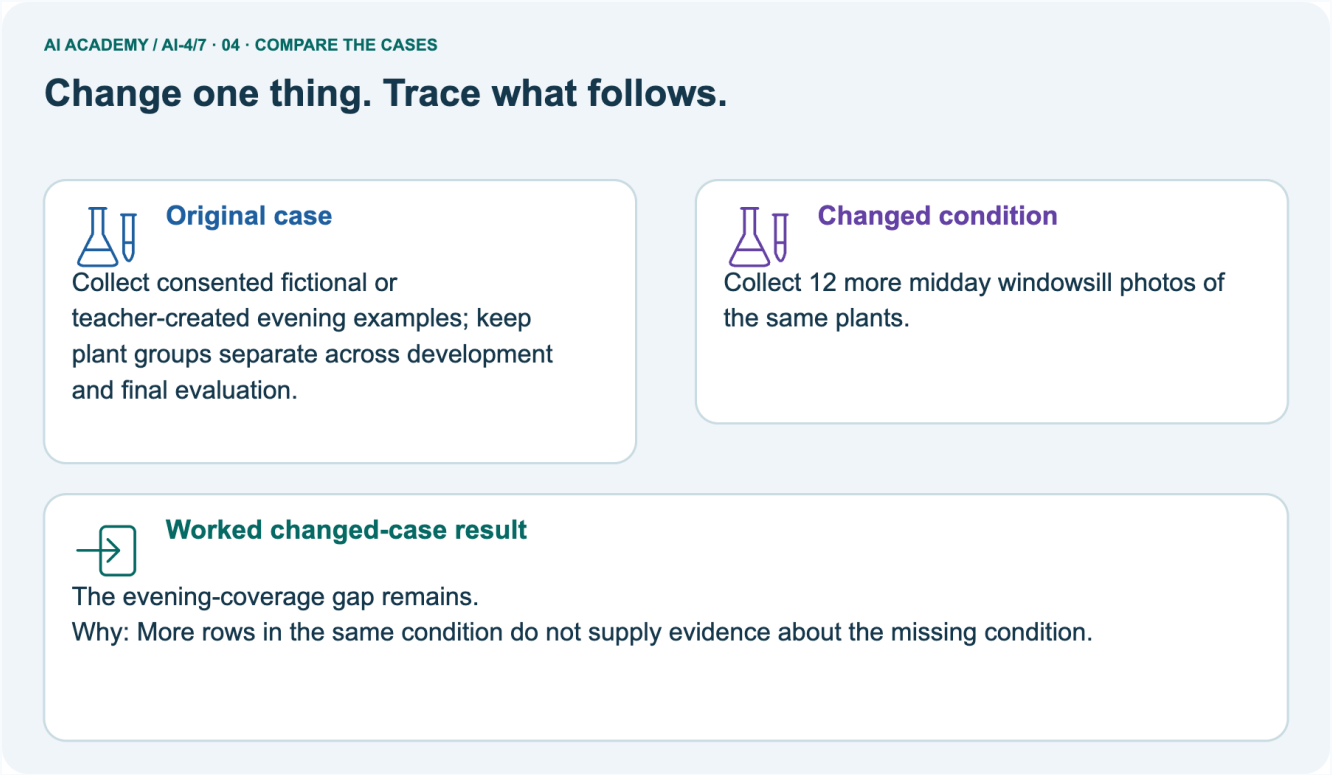
Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict twelve more same-condition views

- Change one thing: Collect 12 more midday windowsill photos of the same plants. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain limits and missing evidence

- Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

- Read the labelled picture: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Review · Earlier trip dataset-card practice

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add **intended use**, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 7 • WORKED EVIDENCE

1. Intended use includes indoor classroom plants under day and evening light.
2. Sample A has 90% daylight photos from one camera.
3. Sample B balances lighting, camera distance, and several plant types.
4. Prefer B for this use, while documenting remaining gaps rather than claiming universal coverage.

STEP 1 **Intended use** includes indoor classroom plants under day and evening light.

STEP 2 **Sample** A has 90% daylight photos from one camera.

STEP 3 **Sample** B balances lighting, camera distance, and several plant types.

STEP 4 Prefer B for this use, while documenting remaining gaps rather than claiming universal coverage.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

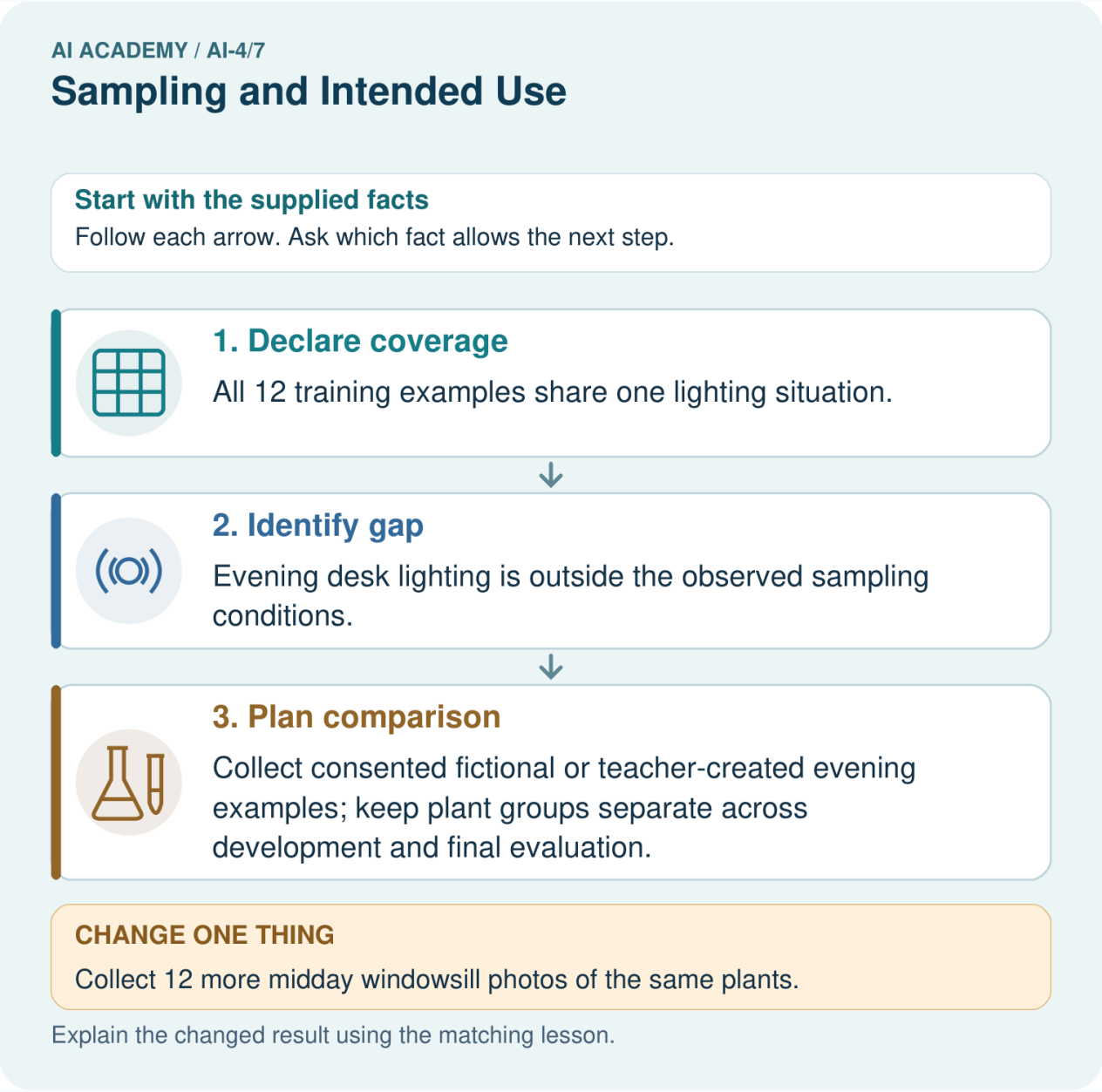
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Intended Use

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Repair sample-size-as-performance reasoning

> Point to the missing cell or missing evaluation result.

Two complete claims to challenge

Claim: B has more photos, so it must have higher model accuracy. The manifests contain no fitted model or predictions; they only show B covers the four declared condition cells more evenly. Another claim: copying every A row ten times fixes camera coverage. It changes row count while camera-B cells remain zero and the plant-ID count remains10.

Correctness of labels and actual performance still need their own evidence.

Propose a revealing changed case

- Diagnose one cause and name the cleanest next experiment:

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A large dataset is representative simply because it contains many examples. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A large dataset is representative simply because it contains many examples.

A closer explanation

A complete-looking card does not prove the table is accurate or suitable for a model. It makes claims and unanswered questions reviewable.

Report raw counts separately from any later accepted subset, with explicit rules for missing, invalid and repeated records. Do not silently change denominators after cleaning.

For this exercise, zero is allowed and unknown is not zero. That is a stated **schema** choice, not a universal rule for all duration datasets.

In a real task, measurement resolution and event definitions determine which values make sense. Dataset inspection also does not replace protecting held out data from model choices.

A card says the dataset is ready because it has no empty cells. Name two checks that claim misses, then write a more accurate statement about Robot practice v1.

MAIN LESSON · SECTION 8

Practise the supplied coverage comparison with less help

> Show counts before choosing a sample or proposing collection.

Current coverage task and response

- Using Coverage v1 A/B manifests, report day/A, day/B, evening/A and evening/B counts for each sample. Explain why B addresses a known gap without proving higher accuracy. If A photos are each copied ten times, state photo and distinct-plant counts and which cells remain missing.

My coverage reasoning and evidence

Explain the pictured gap

- Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Review · Earlier audit practice

A learner replaces both None values with 0 and removes the negative sign from -2. The table now passes a simple nonnegative check, but those edits invent durations.

Keep the original values, record the reason for each flag, and check the source. If a correction is supported, save a new version with the original value, corrected value and reason. If not, keep the uncertainty visible.

Another learner says all robot trips are covered because every row says indoor. A **dataset card** should instead state that outdoor trips are absent and that the data is fictional.

Extra copies of existing rows do not create new conditions or new independent **evidence**.

Use workbook W2 for the connected example “Dataset inspection and cards”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

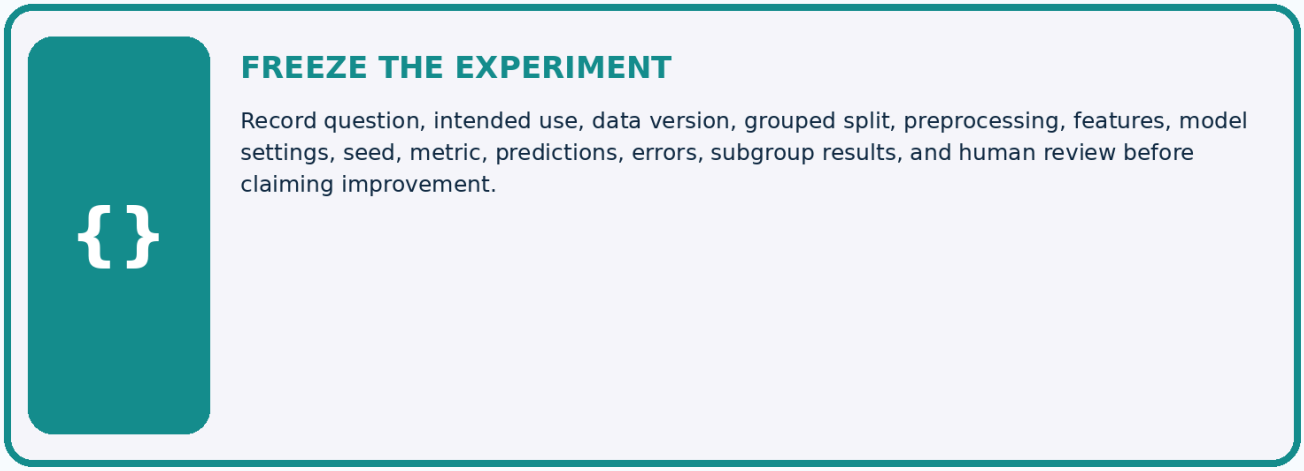
Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + **intended use** + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Interpret the audit counts

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate complete samples against the intended use

> A collection recommendation must name the specific gap it addresses.

Complete normal, sparse, missing and shifted-use probes

Coverage probe	What the supplied evidence supports
Normal: sample B	All four declared light/camera cells have3 photos; no accuracy score.
Boundary: A evening/camera A has1 photo	Nonzero coverage is not proof of adequate evidence or stable performance.
Missing: no camera-B rows in A	State the zero cells and collect planned relevant observations; do not assume failure or success.
Shifted proposal: new outdoor camera C use	Outside the stated day/evening × A/B contract; revise intended use and collect/evaluate relevant new conditions.

Compare, count, plan and record limits

1. Compare both complete manifests, count photo rows and distinct plant IDs, then fill a four-cell coverage table. Choose a targeted collection plan and record unknown species, label balance, selection method and actual-use frequencies. Recount ten copies per A photo to test the more-data claim.

   Describe source-group separation for later evaluation and keep any final evaluation observations separate from development choices. This exercise supplies counts but no fitted model, predictions or measured deployment performance. Reserve the selected outdoor/other-camera task for an individual attempt.

Keep actual or traced evidence

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

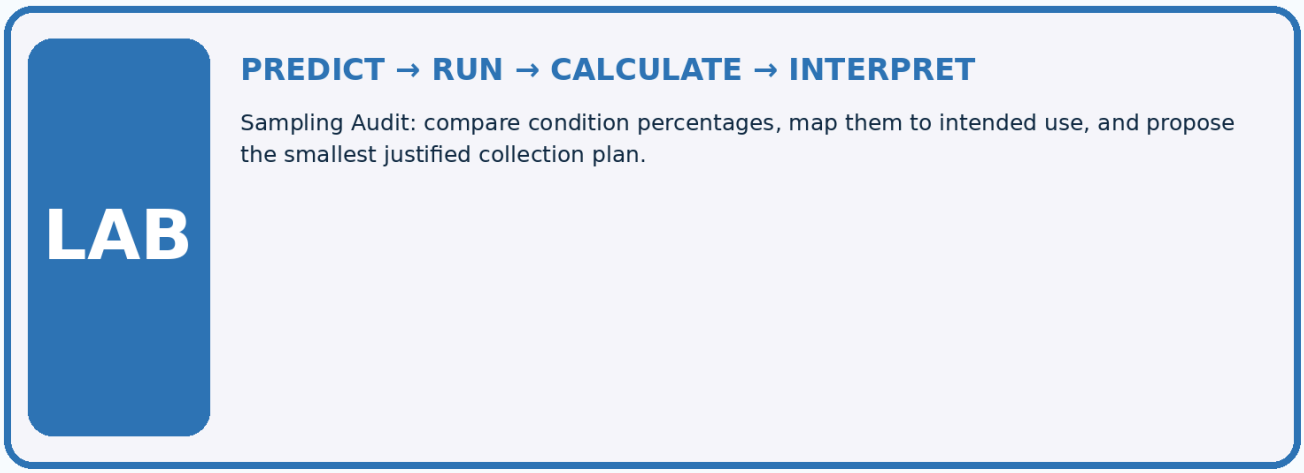
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Sampling Audit: compare condition percentages, map them to **intended use**, and propose the smallest justified collection plan.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: plan checks for outdoor photos and other cameras

> Give a bounded plan using only the supplied facts.

Selected task and scope

Training plant photos are all indoors on one phone. **Intended use** includes outdoor photos on other cameras. Plan a sampling check before making a general accuracy claim.

This selected task asks for a collection/evaluation plan, not an invented score. The source gives no outdoor counts, camera specifications, labels or predictions. Name missing conditions, preserve shared plant IDs when collecting related views, distinguish development collection from fresh final evaluation and state what remains unknown. No particular sample size guarantees adequate accuracy.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Training photographs all show mature plants in bright light, but **intended use**rs photograph seedlings indoors. What claim is unsupported and what data is needed?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Freeze the Plant Classifier intended-use statement and a coverage table for species, lighting, camera, and health class.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Training uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep intended use and coverage beside the project table

> Make the next data decision traceable to a declared condition gap.

Project artifact and limits

THIS WEEK'S CHECKPOINT Freeze the Plant Classifier intended-use statement and a coverage table for species, lighting, camera, and health class.

Freeze an intended-use statement with object types, lighting, camera conditions, users and excluded uses. Attach a coverage table with photo counts, distinct source plants, unknown dimensions and collection priorities. Use the classroom observable label rule rather than implying a diagnosis.

Do not call a coverage table an accuracy evaluation; later fixed-model tests need trustworthy labels, source-separated examples and per-slice denominators.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Record evidence and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

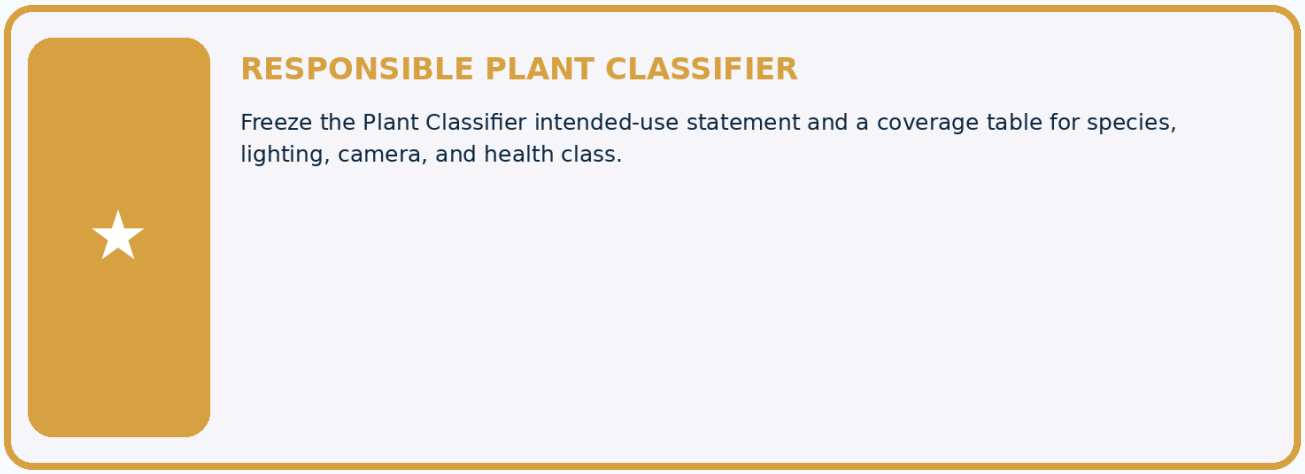
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a condition gap and a justified collection plan

> Say what the sample shows and which performance claim remains untested.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Inspect row meaning **missing value**s and repeated records

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
population	full group the product may encounter	
sample	examples selected from that population	
representation	coverage of relevant conditions	
intended use	specific approved purpose and setting	

Closer-explanation terms

Word	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero

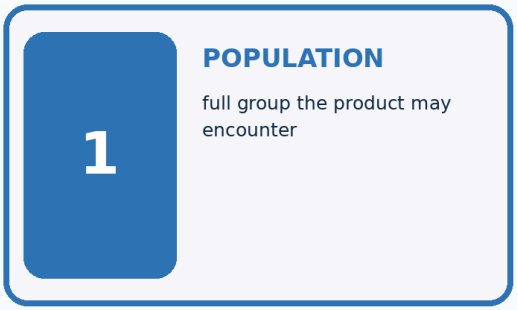
Word	Meaning
Dataset card	A concise record of dataset contents creation intended use and limitations

▼ Earlier wording, alternatives and picture transcripts

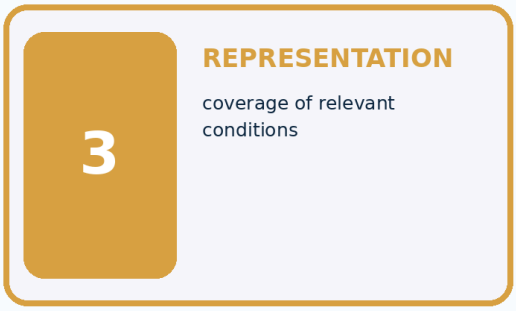
VOCABULARY

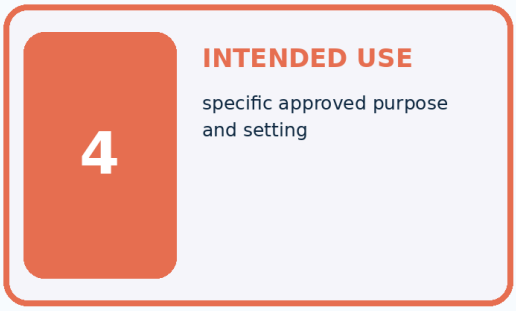
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A large dataset is representative simply because it contains many examples.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **population**, **sample**, **representation**, **intended use**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A large dataset is representative simply because it contains many examples.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State **intended use**, metric, limitation, and human responsibility.

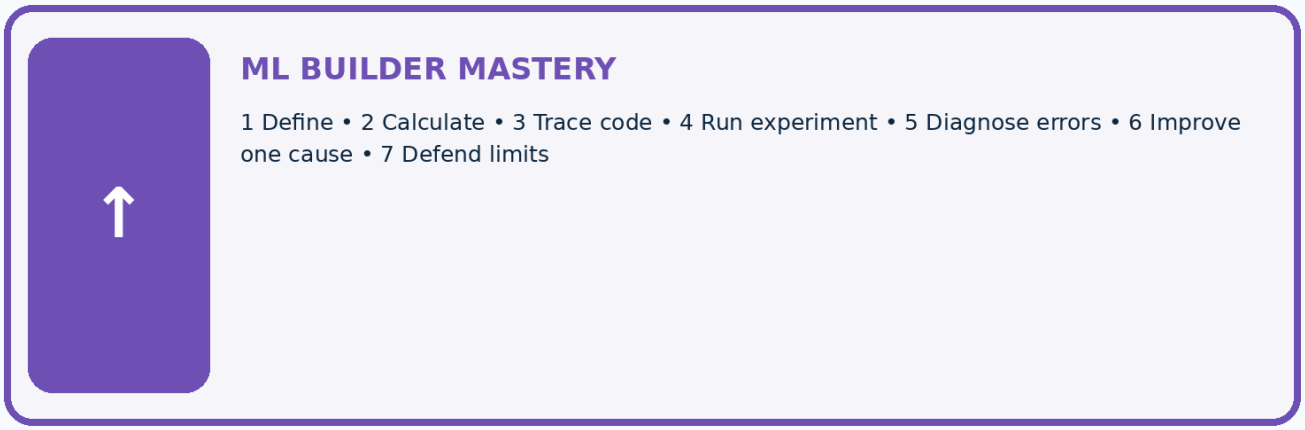
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected outdoor/other-camera collection plan for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

- Independent **evidence** and defense:

- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Compare two **sample**s against a deployment **population** and identify missing conditions.

NEXT BRIDGE Carry the experiment **evidence** into: Leakage.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Training photos use indoor white backgrounds; **intended use** is outdoors.

Identify the coverage gap.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

Evidence or stage	Value or result
Training	Indoor white backgrounds
Intended use	Outdoor plants
Possible shift	Lighting, background and device
Evaluation	Inspect outdoor cases separately

▼ Earlier wording, alternatives and picture transcripts

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Match data coverage to intended use

Training photos use indoor white backgrounds; **intended use** is outdoors.

Identify the coverage gap.

Lighting, background and device conditions may differ. Collect representative examples and evaluate outdoor slices separately.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Inspect row meaning missing values and repeated records	Explain the deciding step, show a trace or test, and state one limit.
Document dataset source schema and limitations	Explain the deciding step, show a trace or test, and state one limit.
Separate observed quality flags from justified corrections	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: **Sample** for the conditions in which the plant classifier will actually be used.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- **Sample** for the conditions in which the plant classifier will actually be used.

- Which labelled step matches this detail? Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.
- 1. Identify gap
- 2. Declare coverage
- 3. Plan comparison
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Plan comparison Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Collect 12 more midday windowsill photos of the same plants. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The evening-coverage gap remains.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The evening-coverage gap remains. More rows in the same condition do not supply **evidence** about the missing condition.

Repair and keep the record

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table **schema**. Short practice: Why should the **missing value** not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

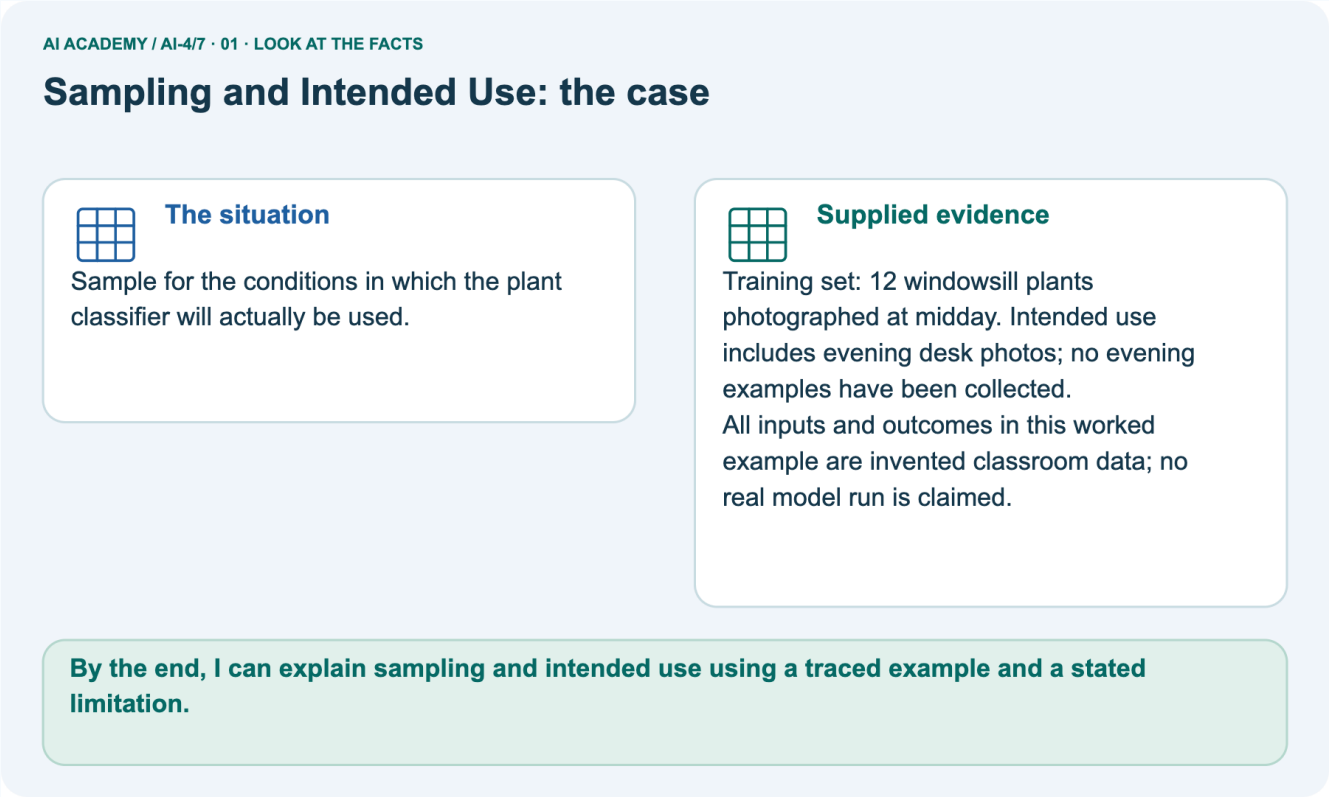
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Sampling and Intended Use: the case



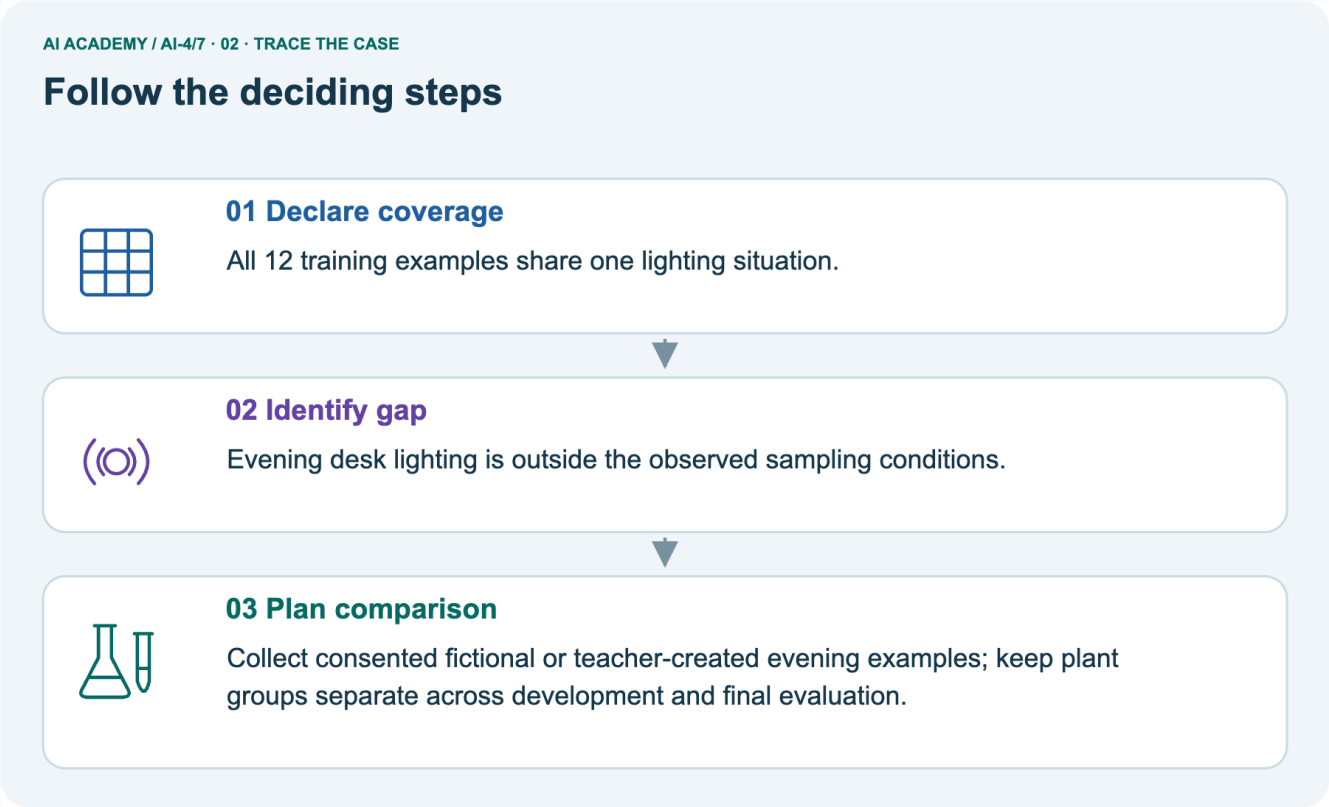
Follow the labelled picture with the matching explanation. Open the vector diagram

Sample for the conditions in which the plant classifier will actually be used. Supplied facts: **Training** set: 12 windowsill plants photographed at midday.

Intended use includes evening desk photos; no evening examples have been collected. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

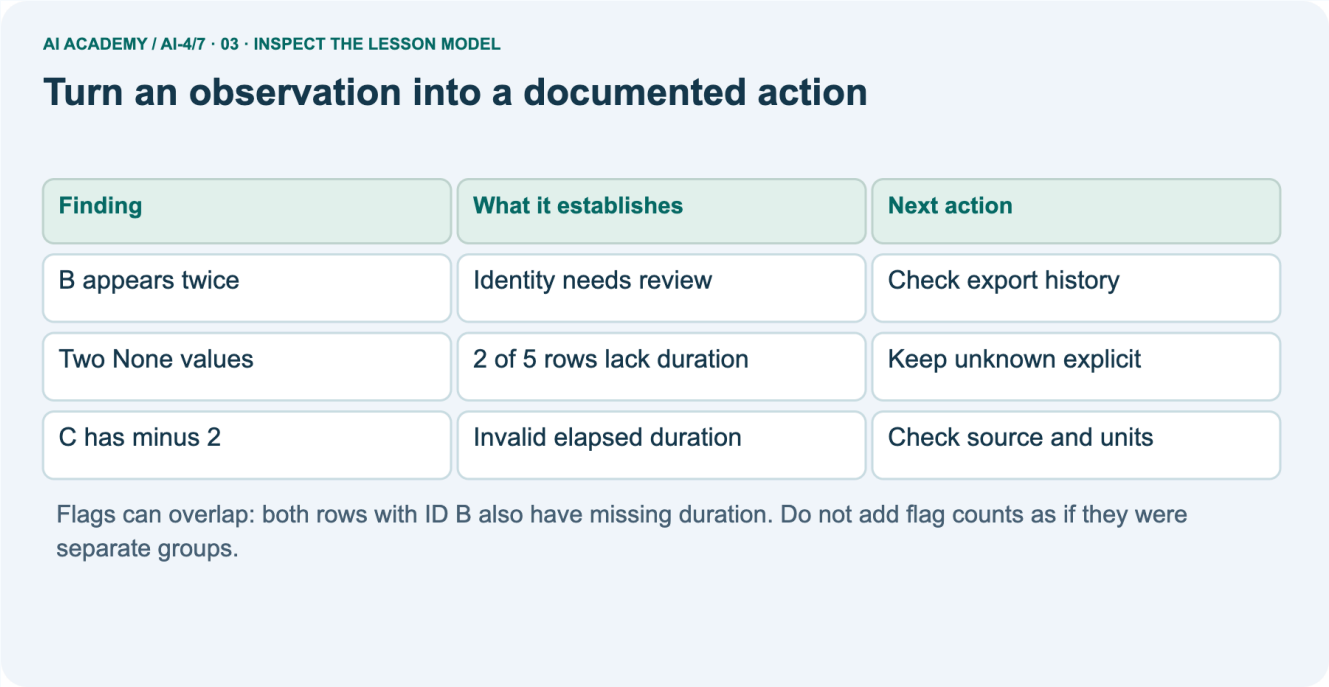


Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coverage: All 12 **training** examples share one lighting situation. → Identify gap: Evening desk lighting is outside the observed sampling conditions. → Plan comparison: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Turn an observation into a documented action



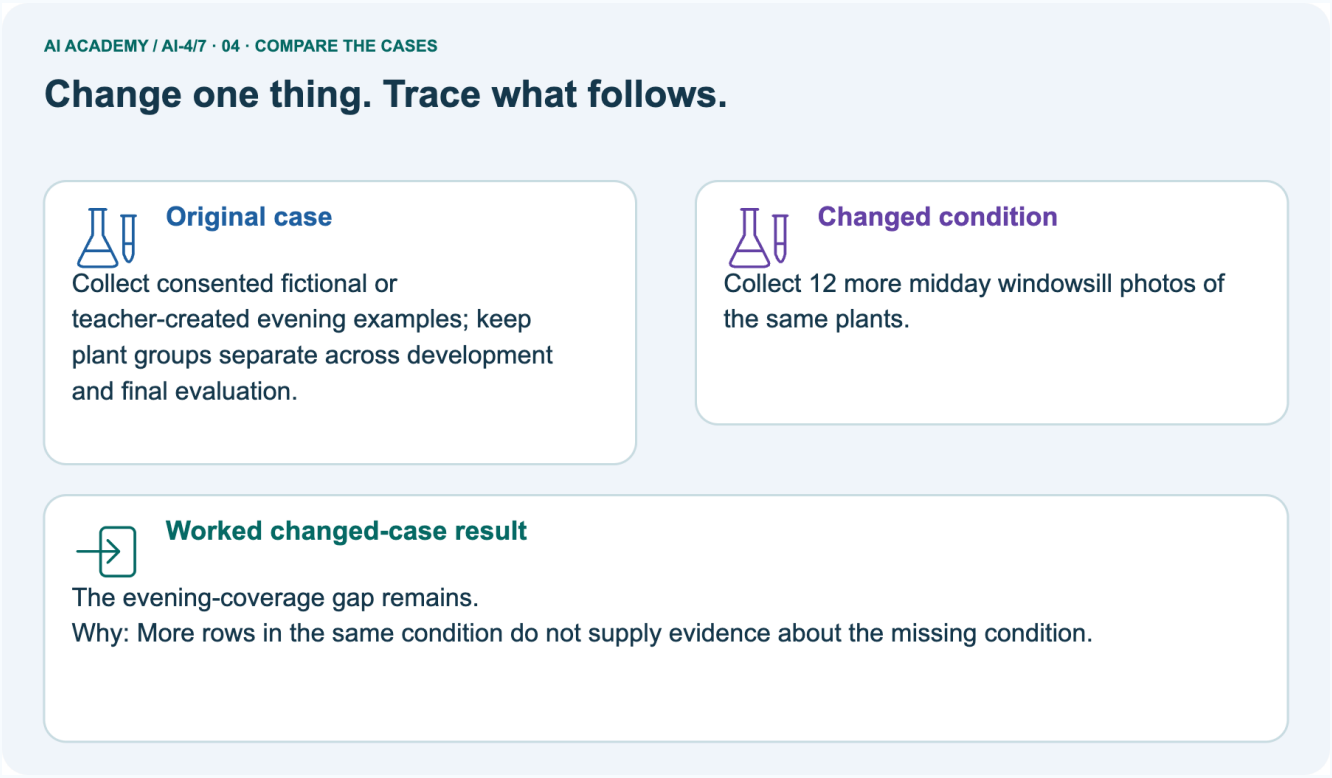
Follow the labelled picture with the matching explanation. Open the vector diagram

Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



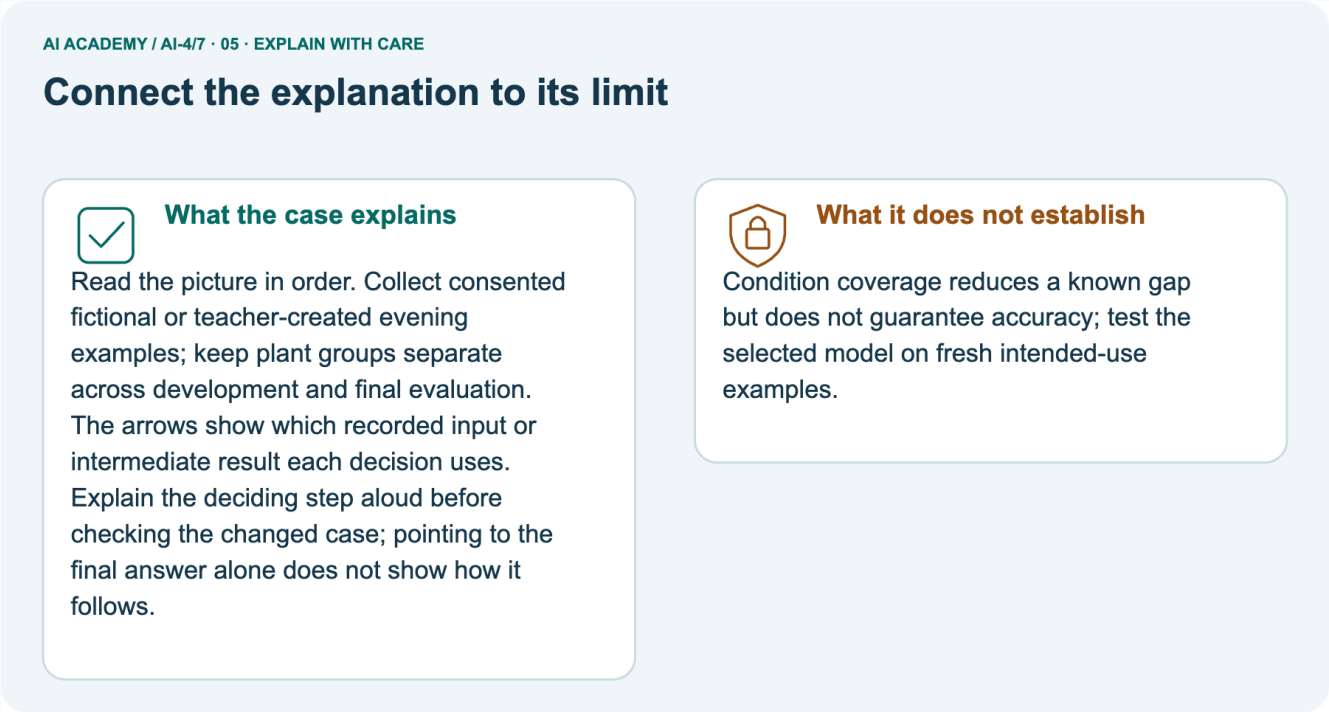
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Changed condition: Collect 12 more midday windowsill photos of the same plants. Expected result: The evening-coverage gap remains. Reason: More rows in the same condition do not supply **evidence** about the missing condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

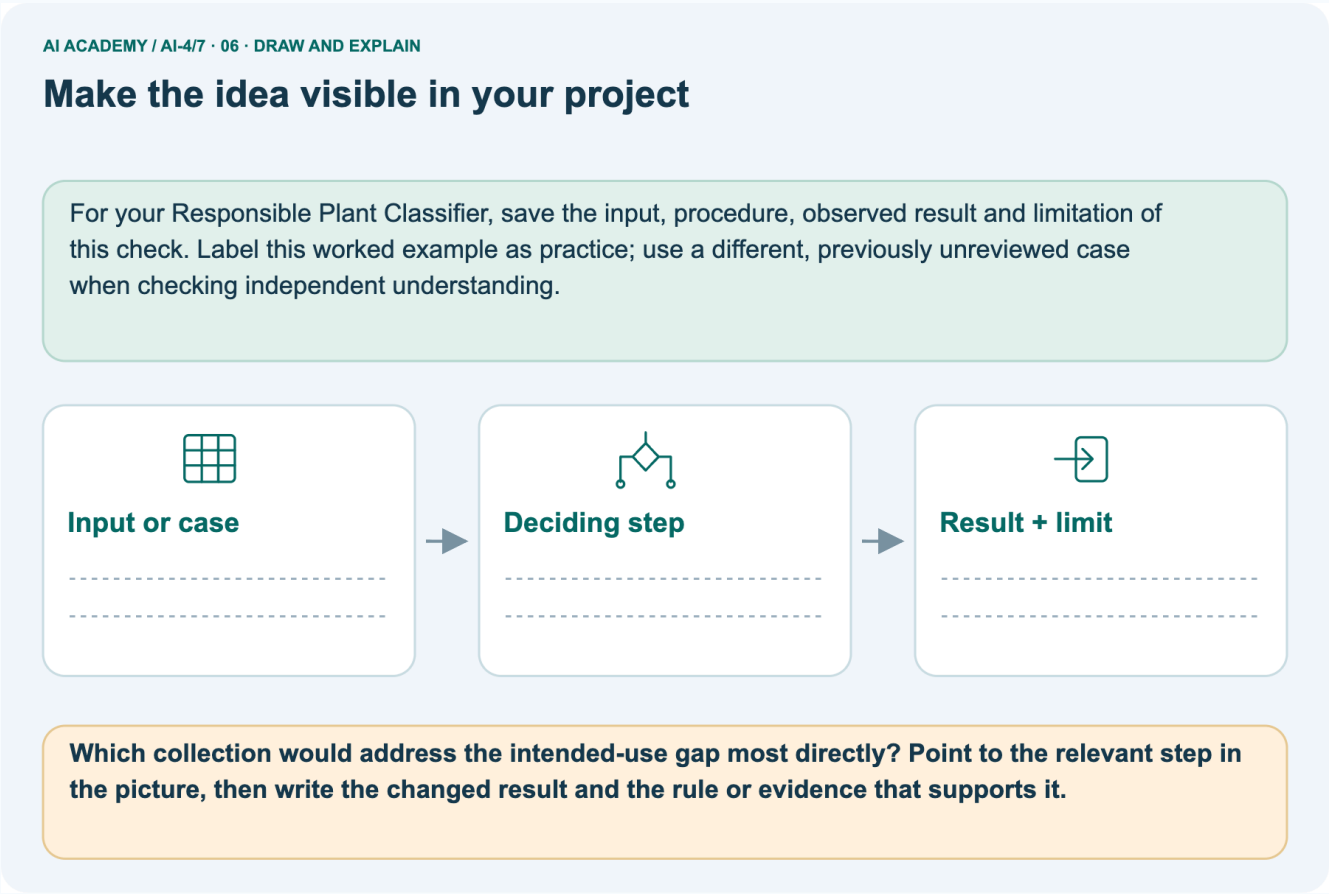
Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sampling and Intended Use: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.